COMSATS UNIVERSITY ISLAMABAD
ATTOCK CAMPUS

Department of Computer Engineering

A Mini Project Report on

UNIVERSITY NETWORK SYSTEM

Design and Simulation of a Secure Campus Network using Cisco Packet Tracer

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Abstract

Reliable connectivity has become central to the day-to-day operation of any modern university: administrative staff, faculty members and students all depend on a network that is fast, always available and, above all, secure. This project presents the design and simulation of a complete campus network for a medium-sized university, built and verified using Cisco Packet Tracer. The proposed network divides the campus into five functional departments — IT, Accounts, Registrar, Faculty and Students — each isolated on its own Virtual LAN (VLAN) so that departmental resources remain private while still allowing controlled inter-department communication through a central routing infrastructure.

The design uses a private addressing scheme (172.16.0.0/12) distributed across the departments, dynamic address assignment (DHCP) for the two largest user groups, and two externally reachable servers (Google and YouTube) that are accessed through an ISP router. The finished network was validated through connectivity (ping) tests and Protocol Data Unit (PDU) simulations both within a single subnet and across different subnets, confirming that the design meets its goal of providing every authorized device with reliable, policy-controlled access to the resources it needs.

1. Introduction

1.1 Motivation

Higher-education institutions increasingly rely on digital infrastructure to manage records, deliver learning material and support day-to-day administration. Tasks that were once handled entirely on paper — registration, fee processing, result compilation, research collaboration — are now carried out over computer networks, which makes the underlying network design a critical piece of a university's infrastructure rather than a background utility. A poorly designed network can leak sensitive administrative data, slow down academic work, or fail entirely under the load of thousands of simultaneous users.

This motivated the present project: to design a campus network that is realistic in scale, sound in its security boundaries, and simple enough to extend as the institution grows.

1.2 Objectives

The specific objectives of this project were to:

- Design a hierarchical campus network covering five representative university departments (IT, Accounts, Registrar, Faculty, Students).
- Segment the departments logically using VLANs so that each department's traffic and resources stay isolated from the others by default.
- Use a private IP addressing scheme (172.16.0.0/12) and allocate an appropriately sized sub-block to each department.
- Automate address assignment with DHCP for the departments with the largest device counts (Faculty and Students).
- Provide controlled external connectivity through an ISP router to representative external services (modelled here as a Google server and a YouTube server).
- Verify the design using ping tests and PDU simulations between devices in the same and in different subnets.

1.3 Scope

The network models two campus buildings — an Administration building housing the IT, Accounts and Registrar departments, and an Academic block housing the Faculty and Student departments. Each department is represented by two end-devices (the first and last PC in that department's numbering scheme) rather than every physical machine, which keeps the simulation tractable in Packet Tracer while still exercising every layer of the design: access-layer switching, VLAN segmentation, inter-VLAN routing, DHCP, and edge/ISP connectivity. The same logical design can be scaled to a larger number of devices per department without changing its structure.

2. Background

Building a secure, efficient campus network draws on a small set of core networking devices and concepts. This section summarizes the ones this project depends on.

2.1 Networking Devices

2.1.1 Cabling

Two twisted-pair cabling standards were used while building the topology. A straight-through cable, in which the pin order is identical at both ends, is used to connect dissimilar devices such as a switch to a router or a PC to a switch. A crossover cable, which swaps the transmit and receive pairs (T568A at one end, T568B at the other), is instead used to connect two similar devices directly, such as router-to-router or switch-to-switch links.

2.1.2 Switches

A switch operates at the data-link layer and forwards frames only to the port associated with the destination MAC address, rather than broadcasting to every port as a hub would. This makes switches the natural choice for connecting end devices within a department, since it keeps unnecessary traffic off the wire and allows multiple conversations to happen on the same switch simultaneously. Managed switches additionally support VLAN configuration, which is essential to this project's department-level isolation.

2.1.3 Routers

A router works at the network layer, inspecting the destination IP address of each packet and forwarding it toward the correct network. Unlike a switch, a router can connect entirely separate networks together and perform functions such as inter-VLAN routing, NAT, and route selection. This project uses three routers in different roles: an ISP-facing router, a border router, and a campus (college) router, each responsible for a different hop between the internal departments and the outside world.

2.2 Networking Concepts

2.2.1 VLANs

A Virtual LAN groups switch ports into separate logical broadcast domains regardless of their physical location, so that devices in one VLAN cannot see broadcast or unicast traffic belonging to another VLAN unless a router explicitly permits it. Using a VLAN per department keeps sensitive traffic — for example, Accounts department data — away from devices in unrelated departments, shrinks the size of each broadcast domain (which improves performance), and lets the network be reorganized by simply reassigning switch ports rather than re-cabling the building.

2.2.2 DHCP

Dynamic Host Configuration Protocol automatically leases an IP address and related configuration (subnet mask, default gateway, DNS server) to a device when it joins the network, removing the need to configure every host by hand. This project applies DHCP to the Faculty and Student departments specifically because they contain by far the largest number of devices (modelled as 250 and 2000 users respectively); manual addressing at that scale would be both slow and error-prone.

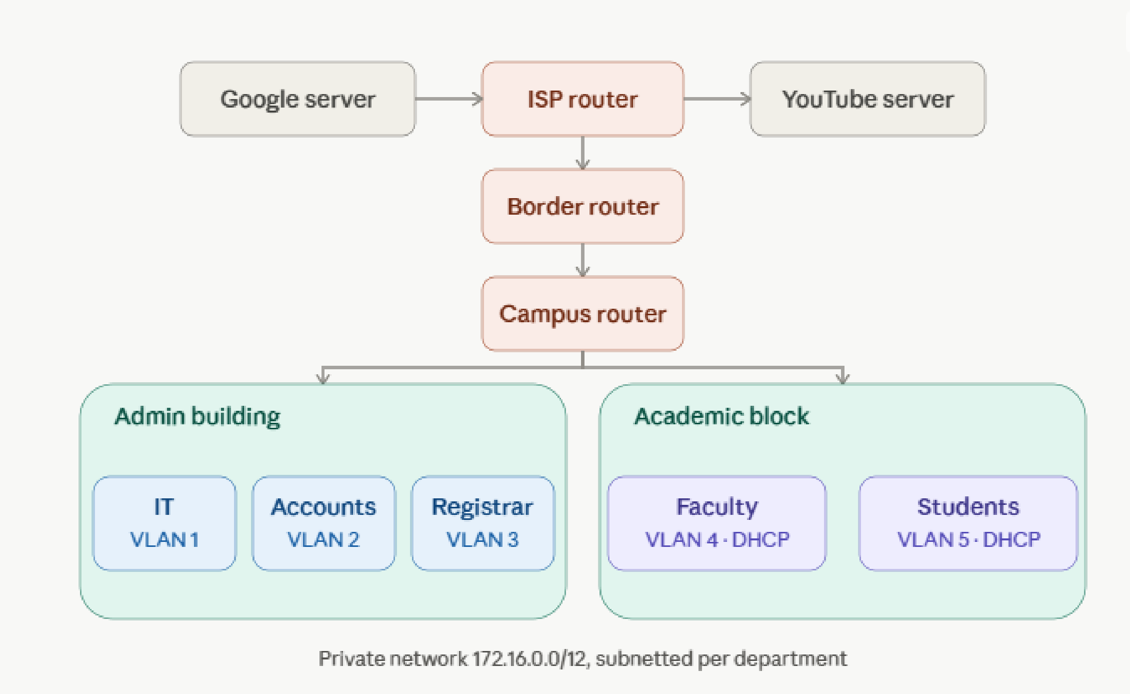

Figure 1: Campus Network Topology with VLAN Segmentation and External Connectivity

2.2.3 Internet Service Provider (ISP) Connectivity

An ISP supplies the physical and logical link between a private network and the public Internet. In this design, the ISP router is the single point through which all traffic destined for external resources — represented in the simulation by a Google server and a YouTube server — must pass, which also makes it a natural point at which to apply perimeter security controls.

2.2.4 Private IP Addressing

Private address ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16, and the carrier-grade NAT range 100.64.0.0/10) are reserved for use inside local networks and are not routed on the public Internet. Because they never appear directly on the Internet, a device using a private address cannot be reached from outside without an intervening NAT translation, which gives the internal network a first layer of protection against unsolicited external traffic. This project uses the 172.16.0.0/12 block for the entire campus, subnetted per department.

2.2.5 Gateways

A default gateway is the device — typically a router interface — that a host sends traffic to whenever the destination lies outside its own subnet. Every department in this design points its end devices at the appropriate departmental gateway, and it is this chain of gateways that ultimately carries traffic up through the college router, the border router and the ISP router to reach an external destination.

3. Methodology / Implementation

The network was designed top-down — first the logical topology and addressing plan, then the device-by-device configuration — and built entirely in Cisco Packet Tracer, which allowed each stage to be tested before moving to the next.

3.1 Design Approach

- Identify the departments to be modelled and assign each one its own VLAN and IP sub-block within 172.16.0.0/12.
- Lay out two buildings — Administration (IT, Accounts, Registrar) and Academic (Faculty, Students) — connected through a campus router.
- Place a border router between the campus router and the ISP router to separate internal routing from Internet-facing routing.
- Configure DHCP scopes for the Faculty and Student VLANs, and static addressing for the smaller departments.
- Add the two external servers behind the ISP router to model Internet-hosted resources.
- Verify every link and VLAN with ping tests and simulated PDUs before considering the build complete.

3.2 Logical Topology

The overall topology, showing how the departments, switches, routers and servers are interconnected, is shown below.

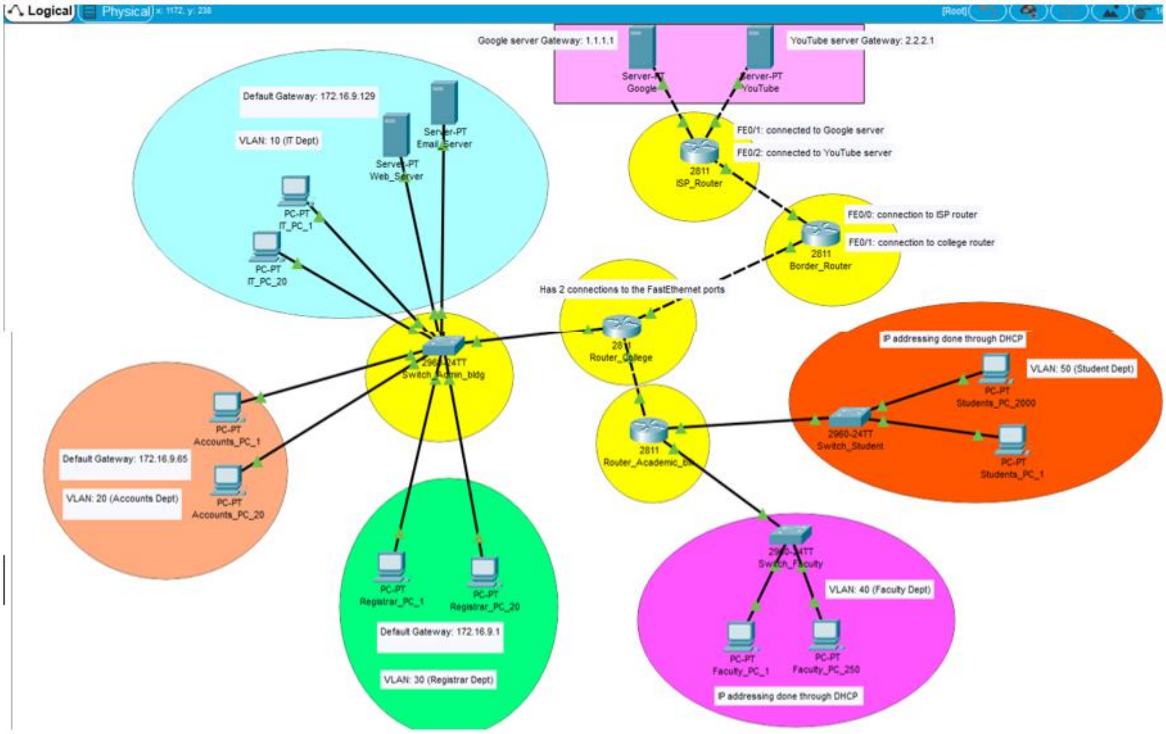

Figure 3.2 — Logical view of the university network

3.3 Server Configuration

3.3.1 Google Server

The Google server was configured behind the ISP router to represent an external web service that all departments can reach through the campus's outbound routing path.

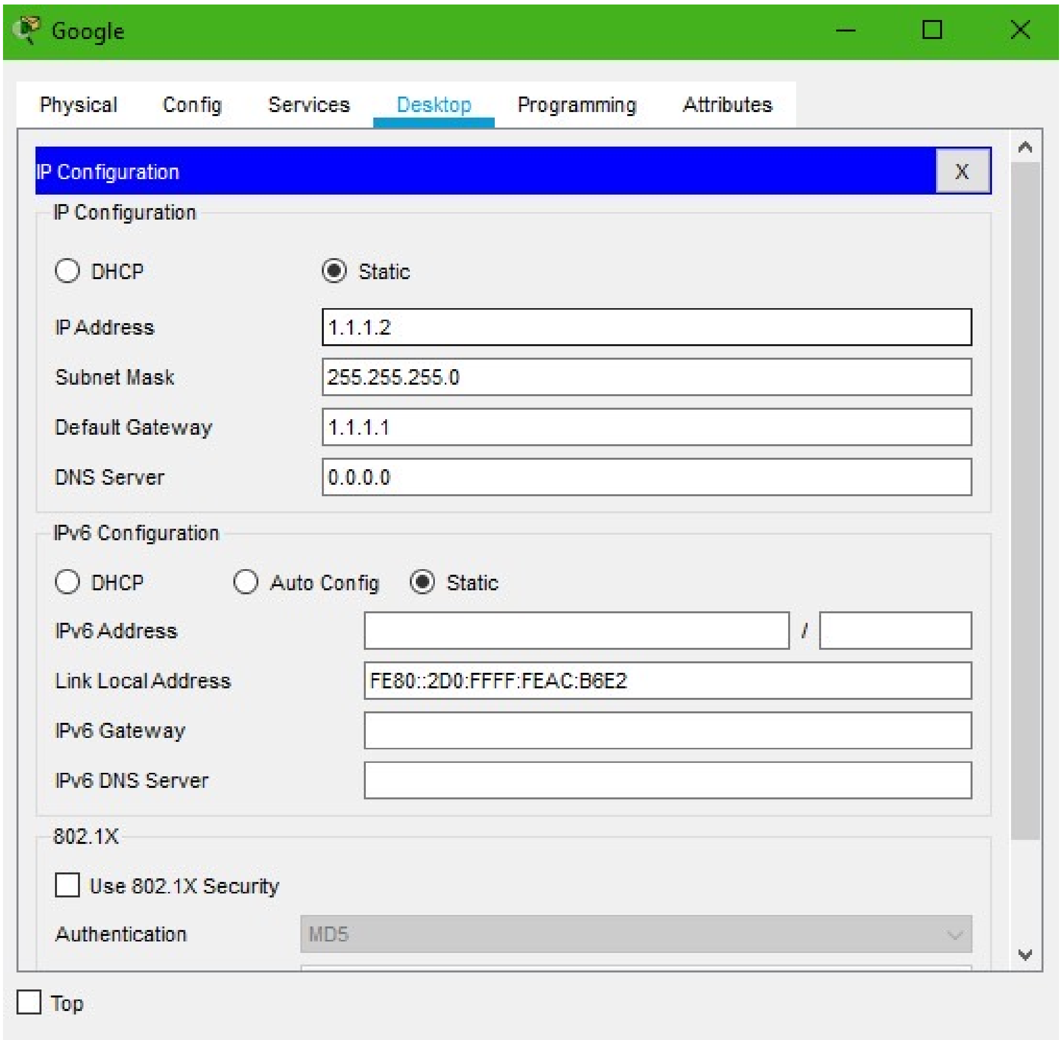

Figure 3.3 — Google server configuration

3.3.2 YouTube Server

A second external server, representing YouTube, was configured alongside the Google server to confirm that the ISP router can correctly route traffic to more than one external destination.

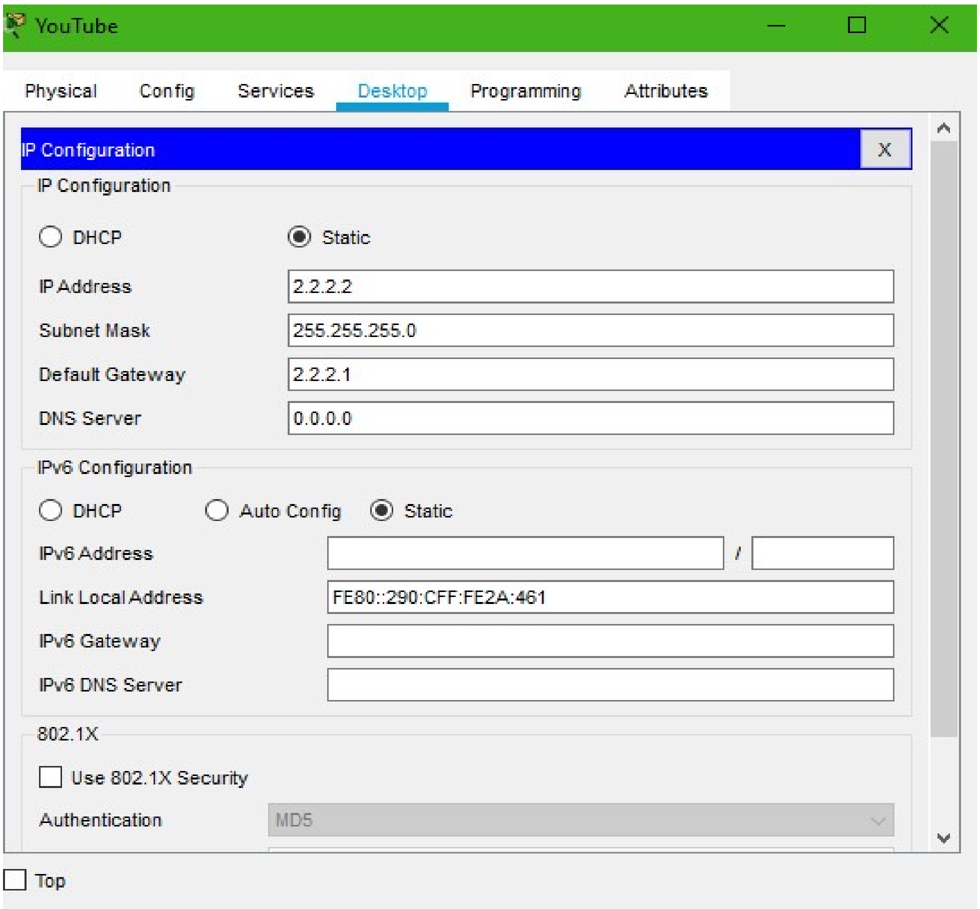

Figure 3.4 — YouTube server configuration

3.4 ISP Router Configuration

3.4.1 ISP Router — Internet-facing Interface

The ISP router's outward-facing interface was addressed and enabled to model the link out to the wider Internet.

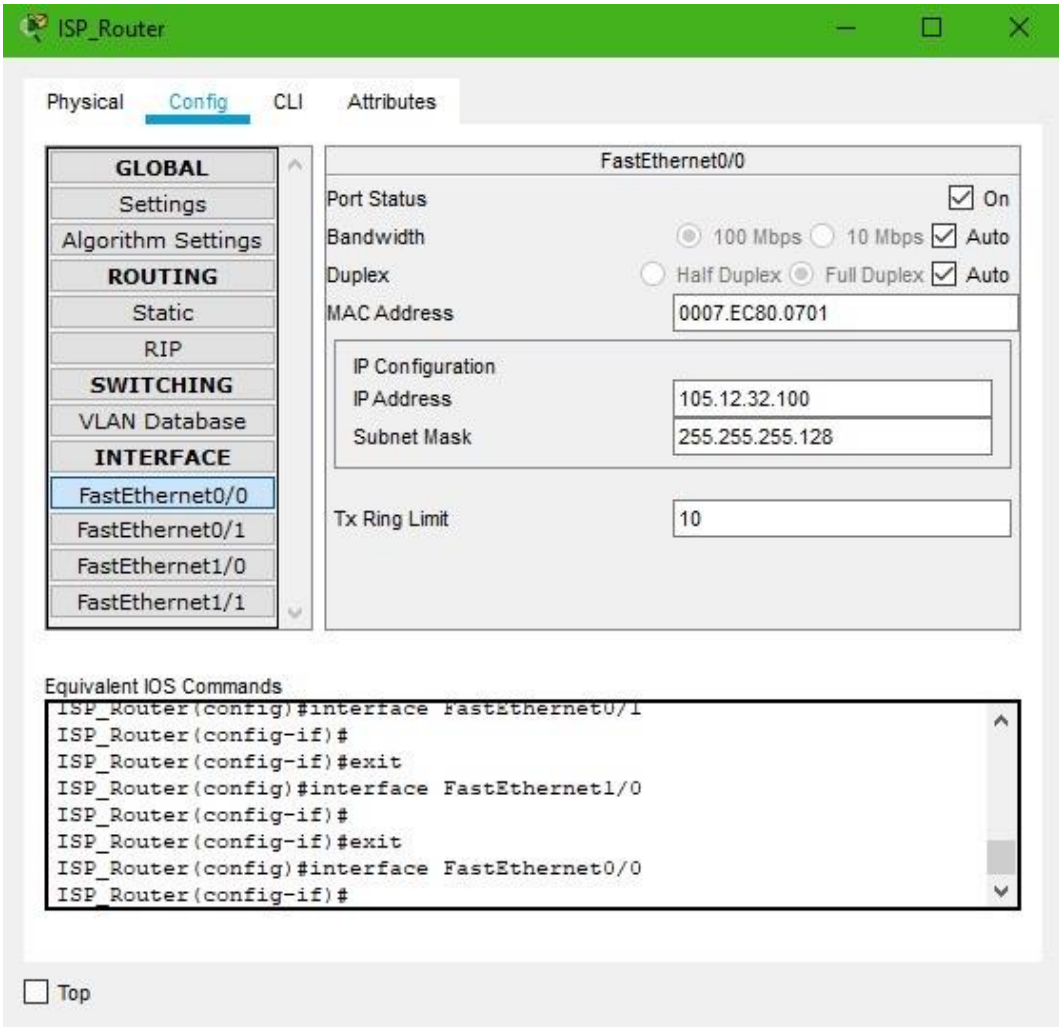

Figure 3.5 — ISP router configuration (Internet side)

3.4.2 ISP Router — Link to Google Server

A second interface on the ISP router was configured to reach the Google server subnet, completing the path between the campus and its external resources.

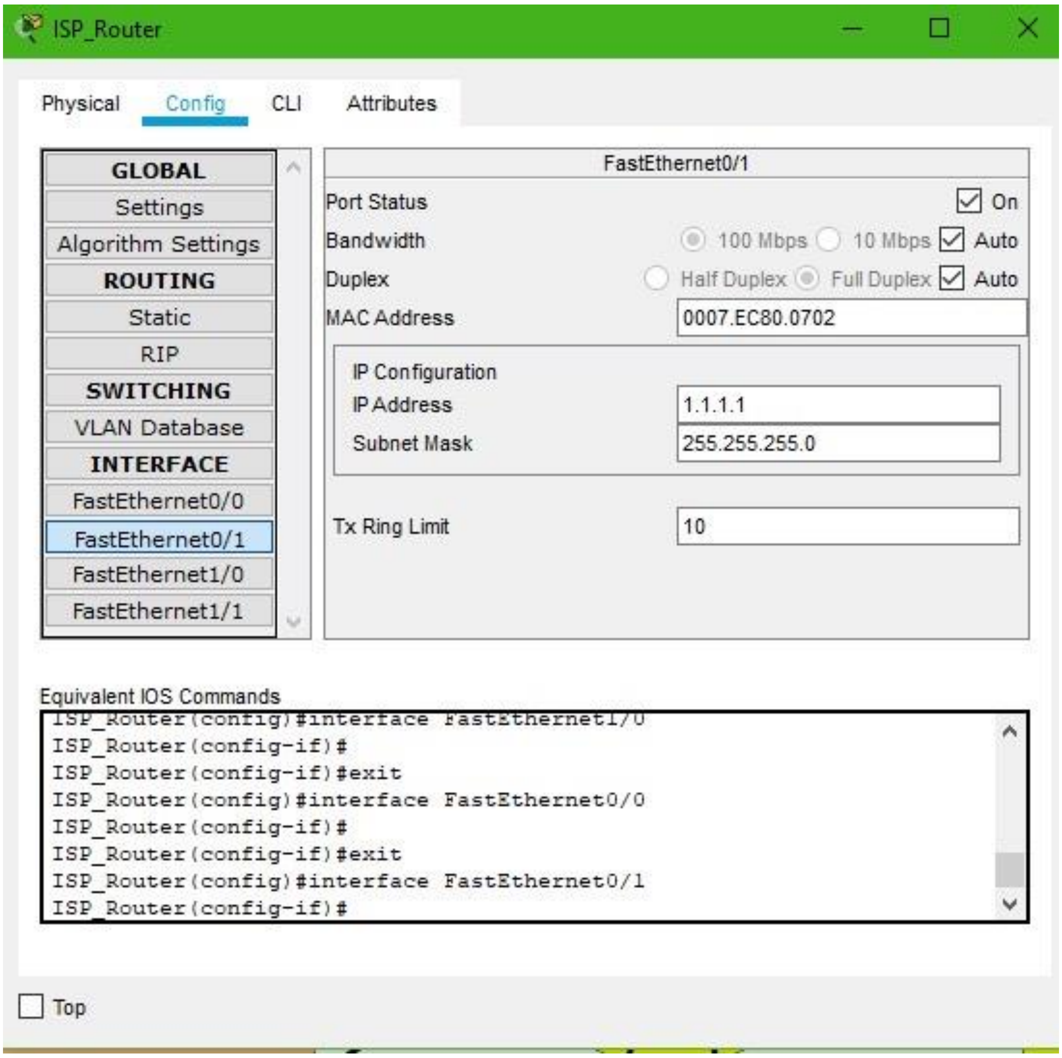

Figure 3.6 — ISP router configuration (server side)

3.5 Border Router Configuration

3.5.1 Border Router — Link to ISP

The border router's upstream interface was configured to connect to the ISP router, forming the campus's single exit point to the outside network.

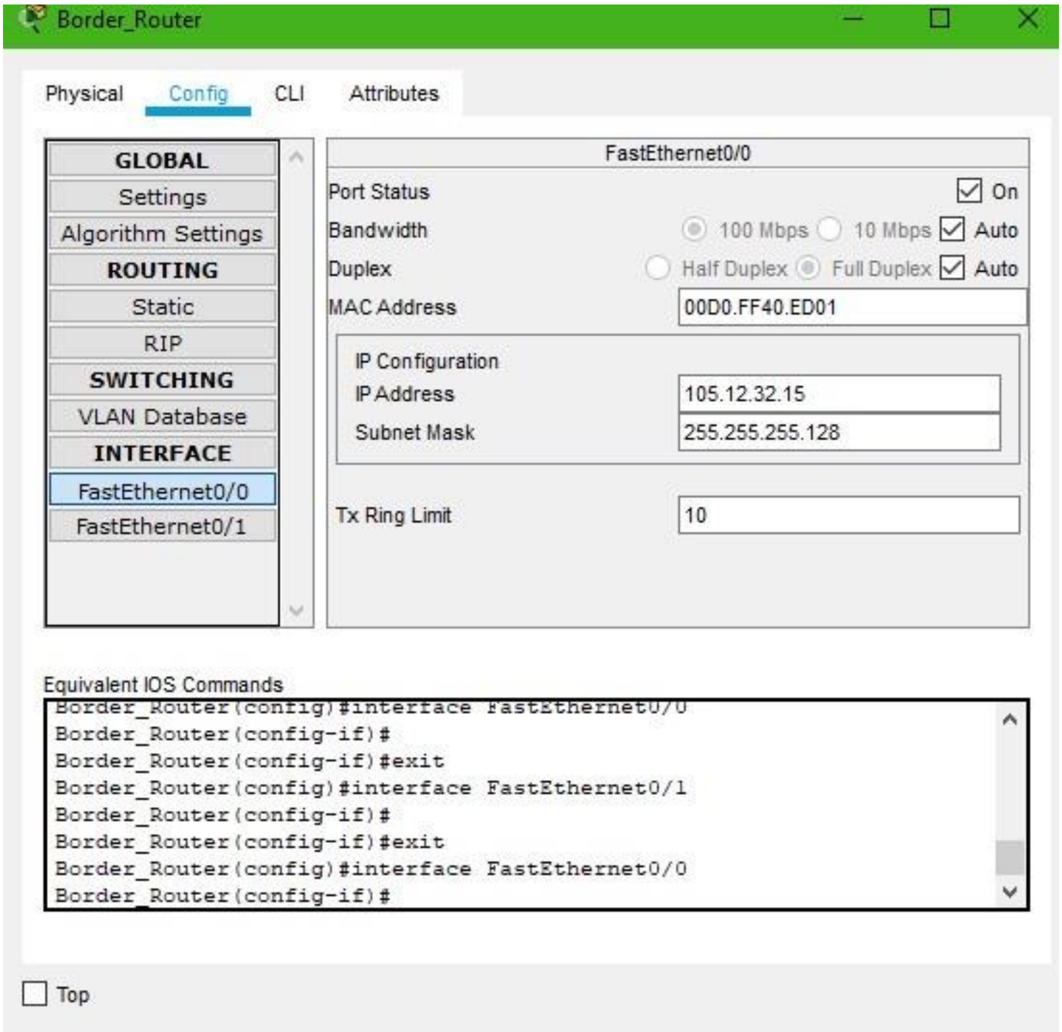

Figure 3.7 — Border router configuration (ISP side)

3.5.2 Border Router — Link to College Router

The downstream interface of the border router was configured to connect to the campus (college) router, extending the routed path into the university's internal network.

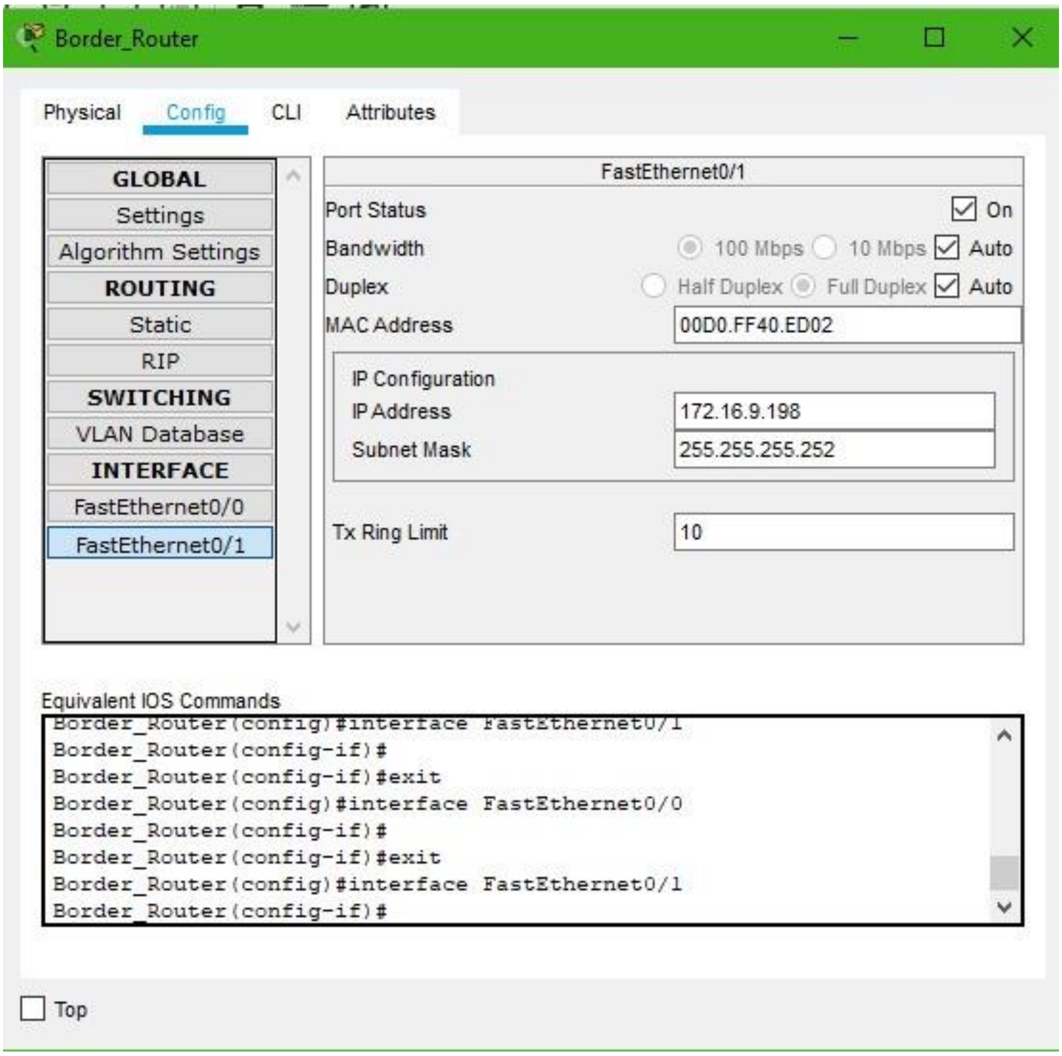

Figure 3.8 — Border router configuration (campus side)

3.6 Campus Router Configuration

3.6.1 Campus Router — Link to Admin Building Switch

The campus router was configured with a sub-interface for the Administration building, enabling inter-VLAN routing for the IT, Accounts and Registrar departments.

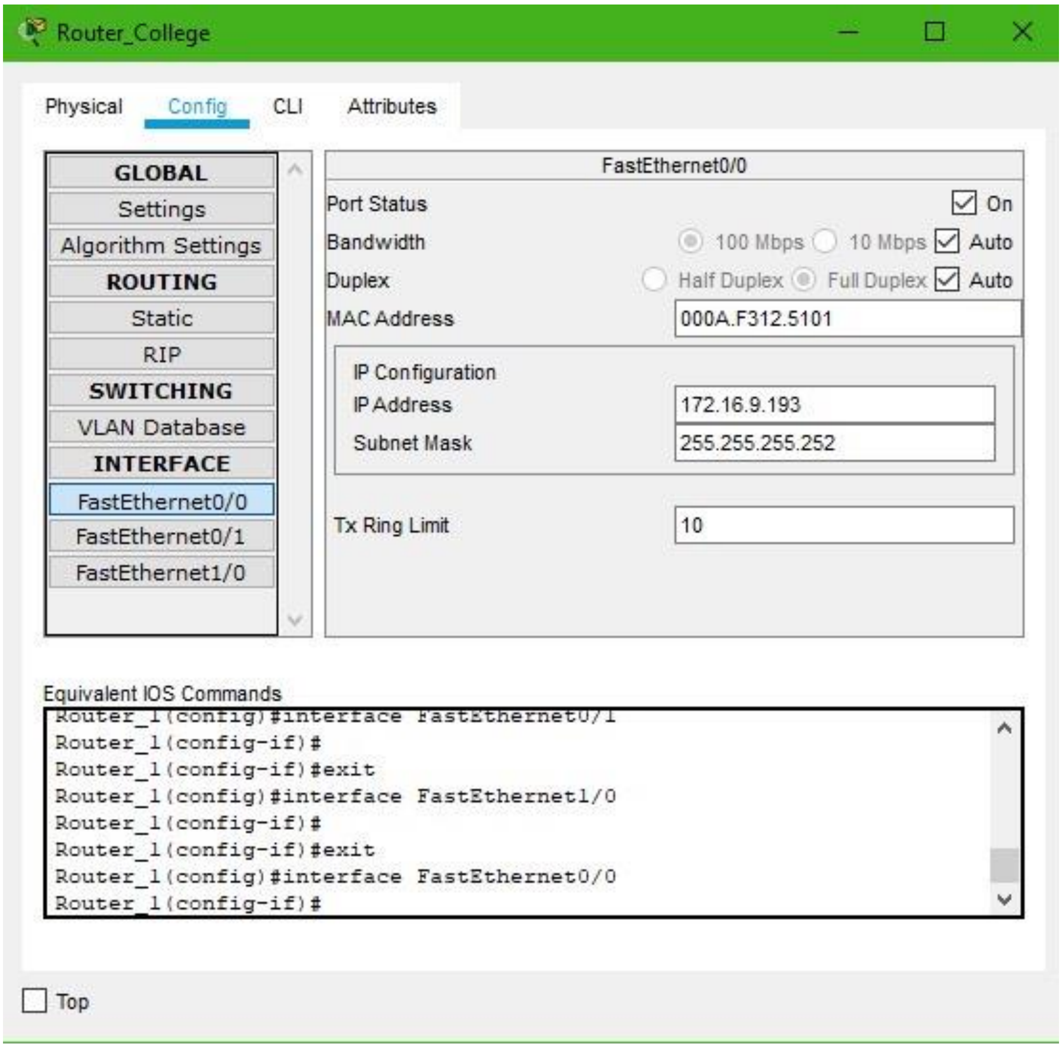

Figure 3.9 — Campus router configuration (Admin building)

3.6.2 Campus Router — Link to Academic Block

A second sub-interface was configured for the Academic block, enabling inter-VLAN routing for the Faculty and Student departments.

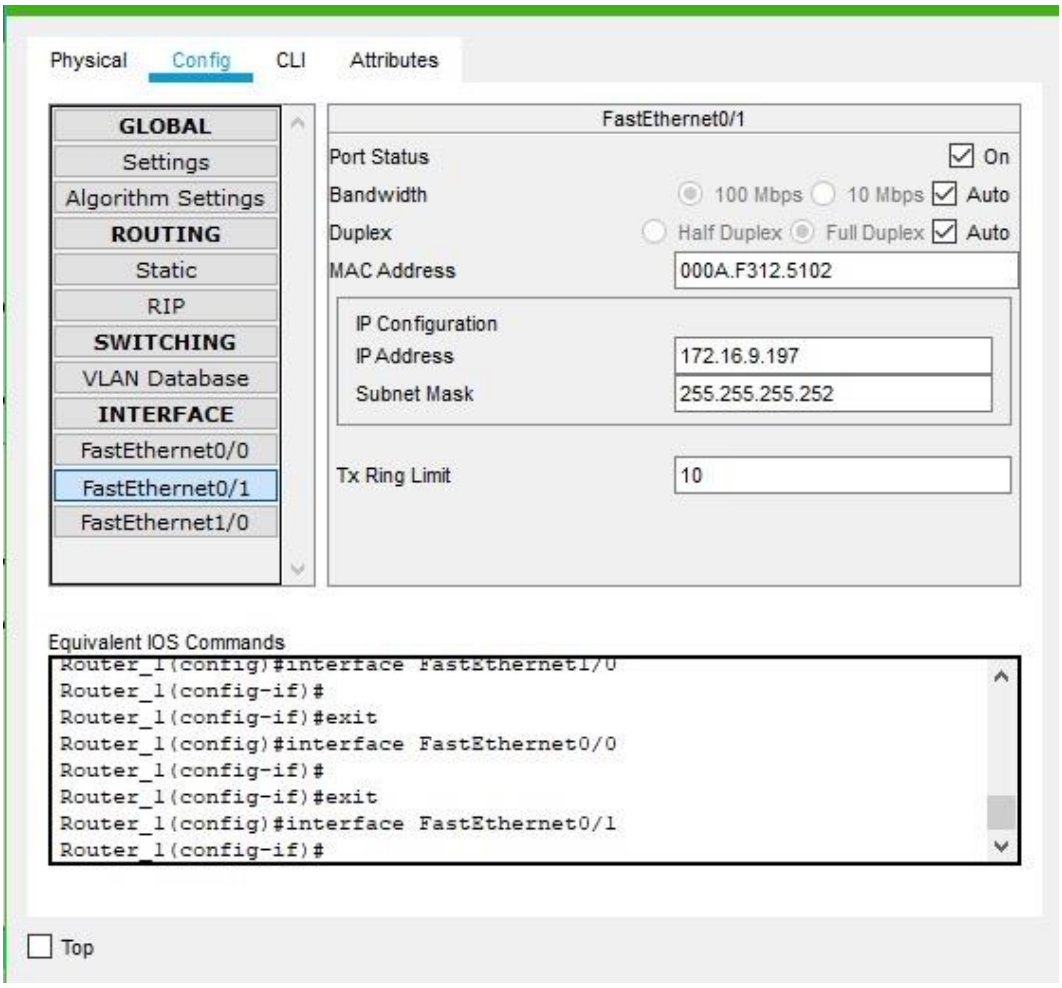

Figure 3.10 — Campus router configuration (Academic block)

3.7 Admin Building Switch — VLAN Configuration

The access switch in the Administration building was configured with three VLANs — one each for IT, Accounts and Registrar — and the corresponding ports were assigned accordingly.

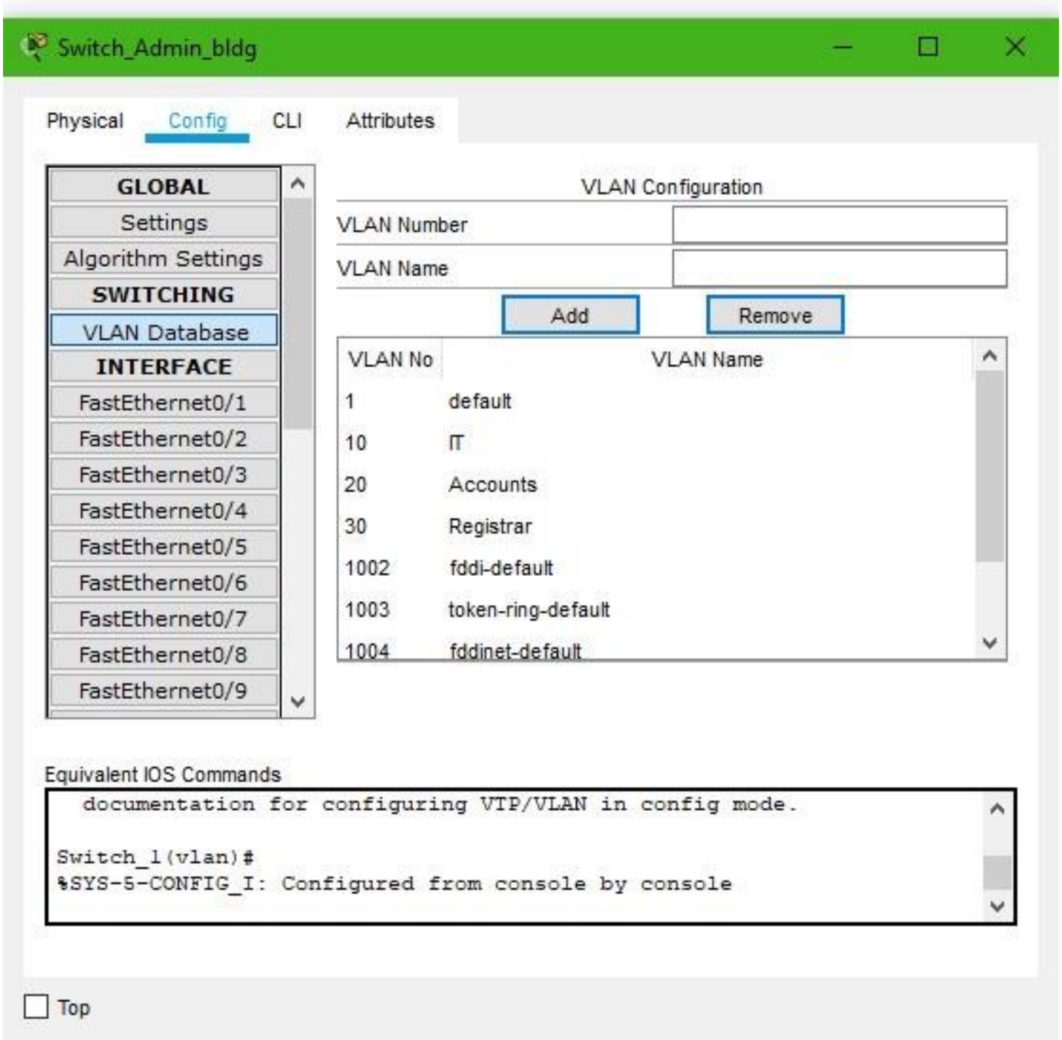

Figure 3.11 — **Admin building switch VLAN configuration**

3.8 IT Department Configuration

3.8.1 Email Server

An internal email server was placed in the IT department to model an intranet service available only to authenticated campus users.

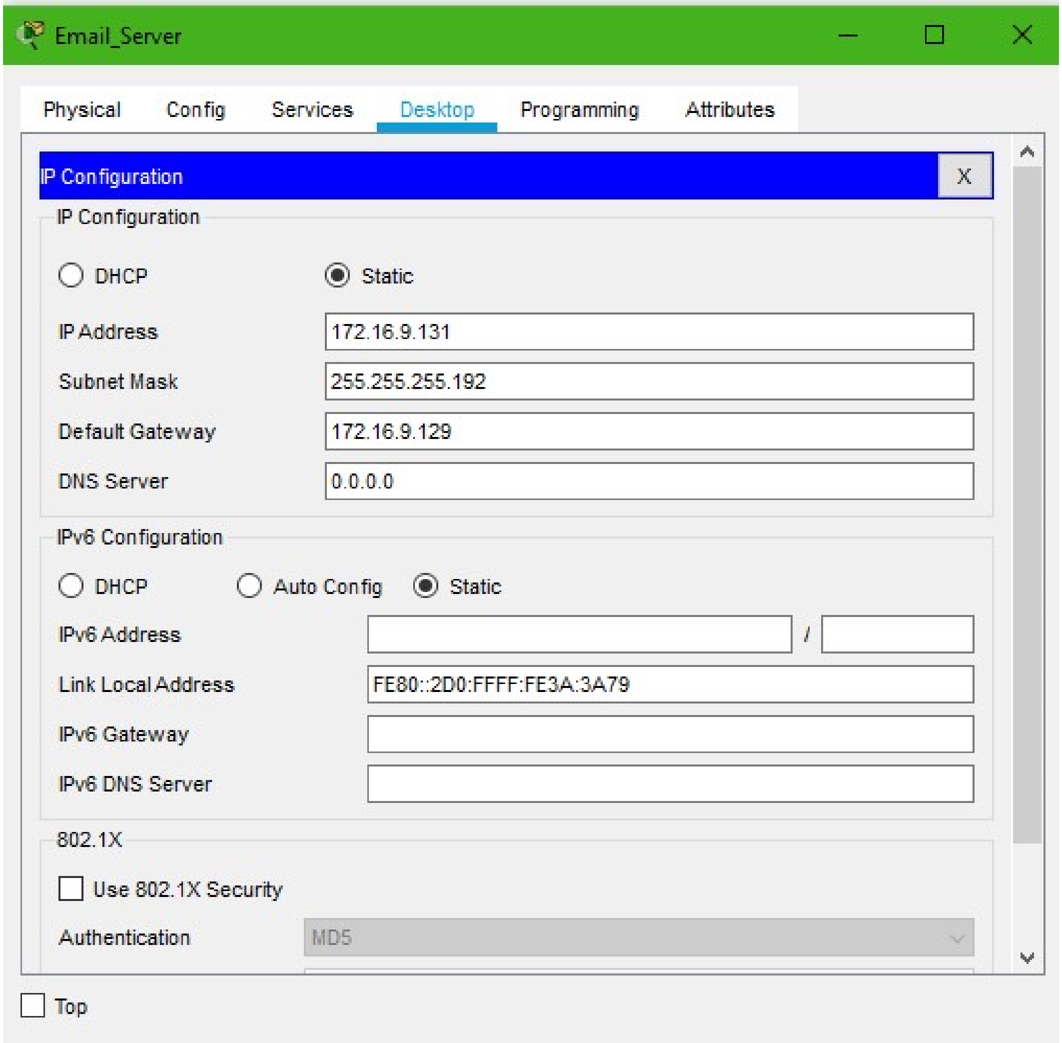

Figure 3.12 — Email server configuration

3.8.2 IT Department End Devices

The IT department's PCs were statically addressed within their VLAN's sub-block and configured with the departmental gateway.

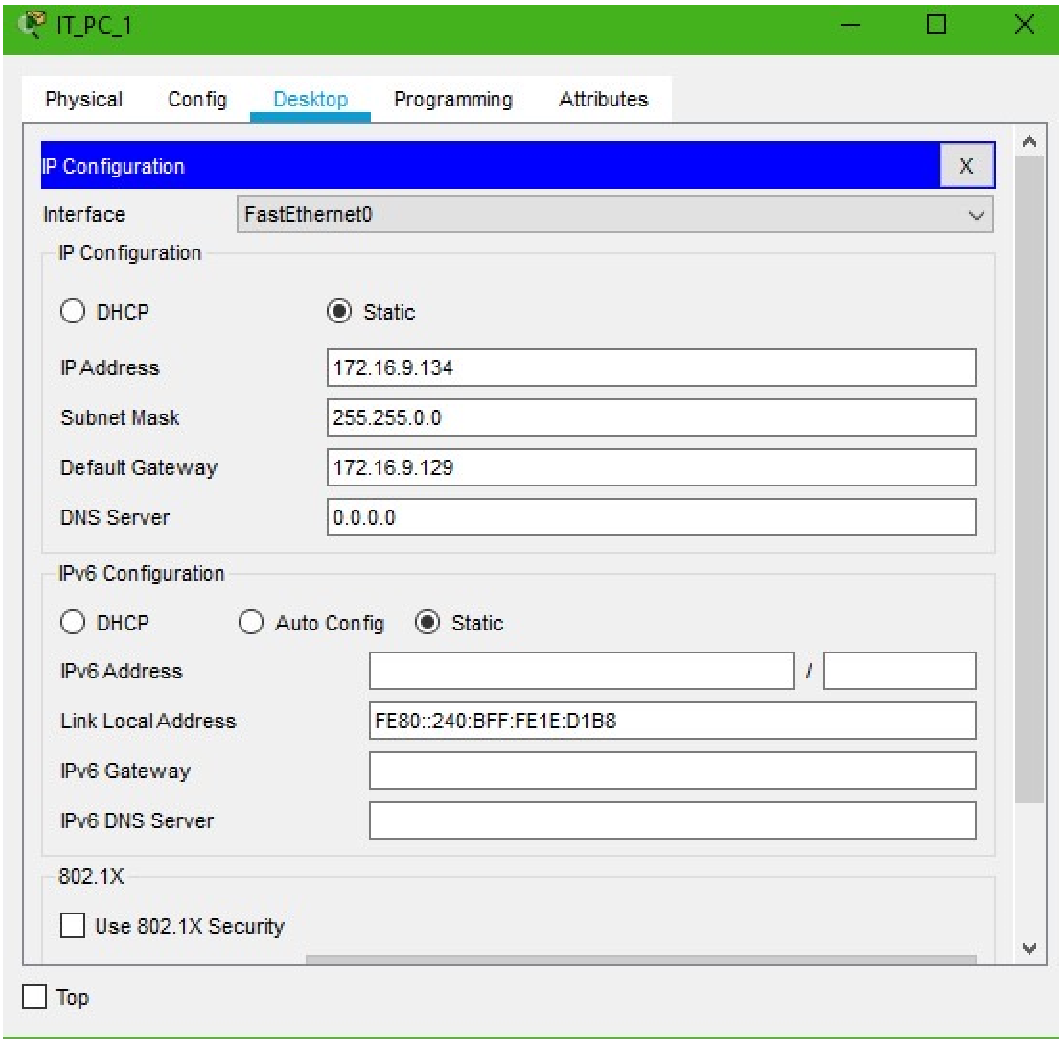

Figure 3.13 — IT department PC configuration

3.9 Accounts Department Configuration

The Accounts department's end devices were configured with static IP addresses inside their VLAN, reflecting the department's smaller, more sensitive user base where manual control over addressing was preferred over DHCP.

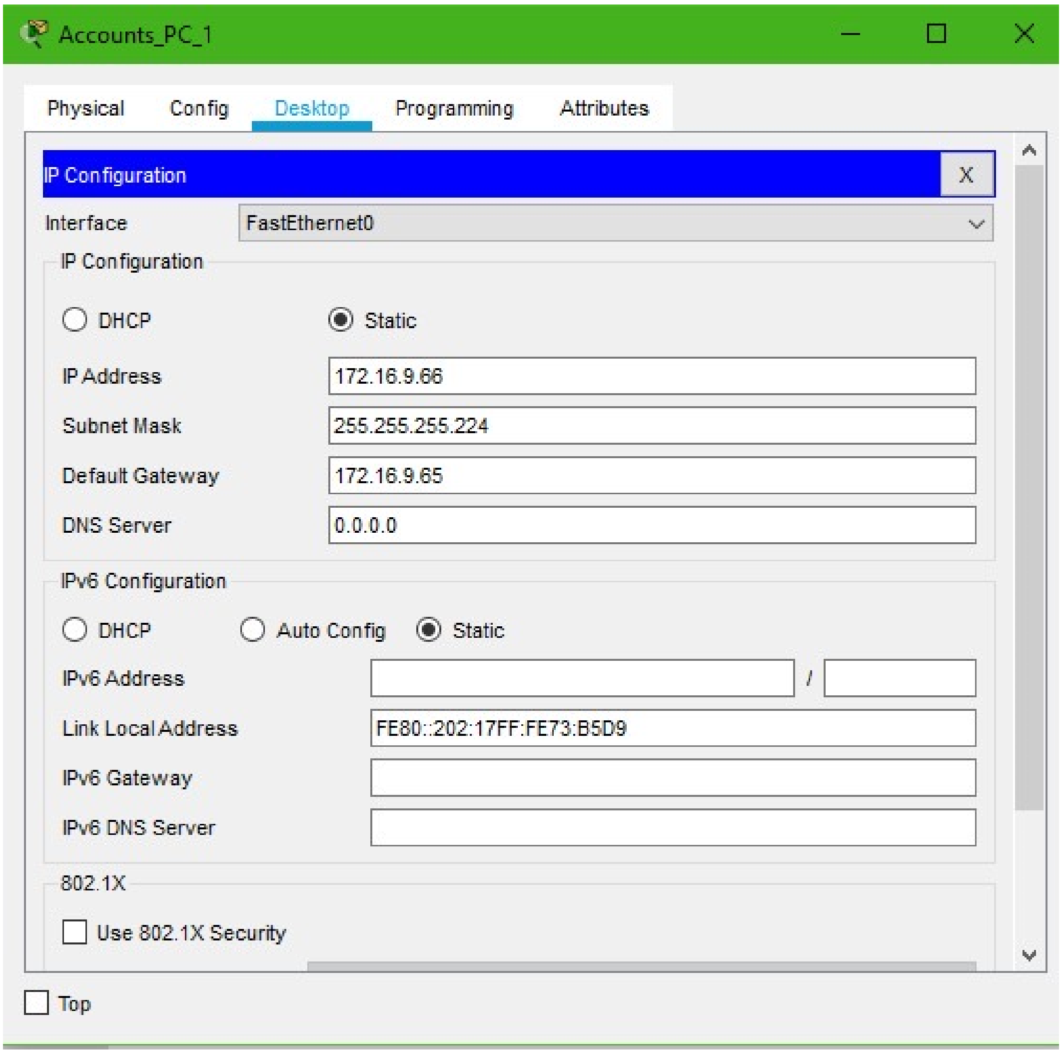

Figure 3.14 — Accounts department PC configuration

3.10 Registrar Department Configuration

Similarly, the Registrar department's PCs were statically addressed within their own VLAN and pointed at the departmental gateway.

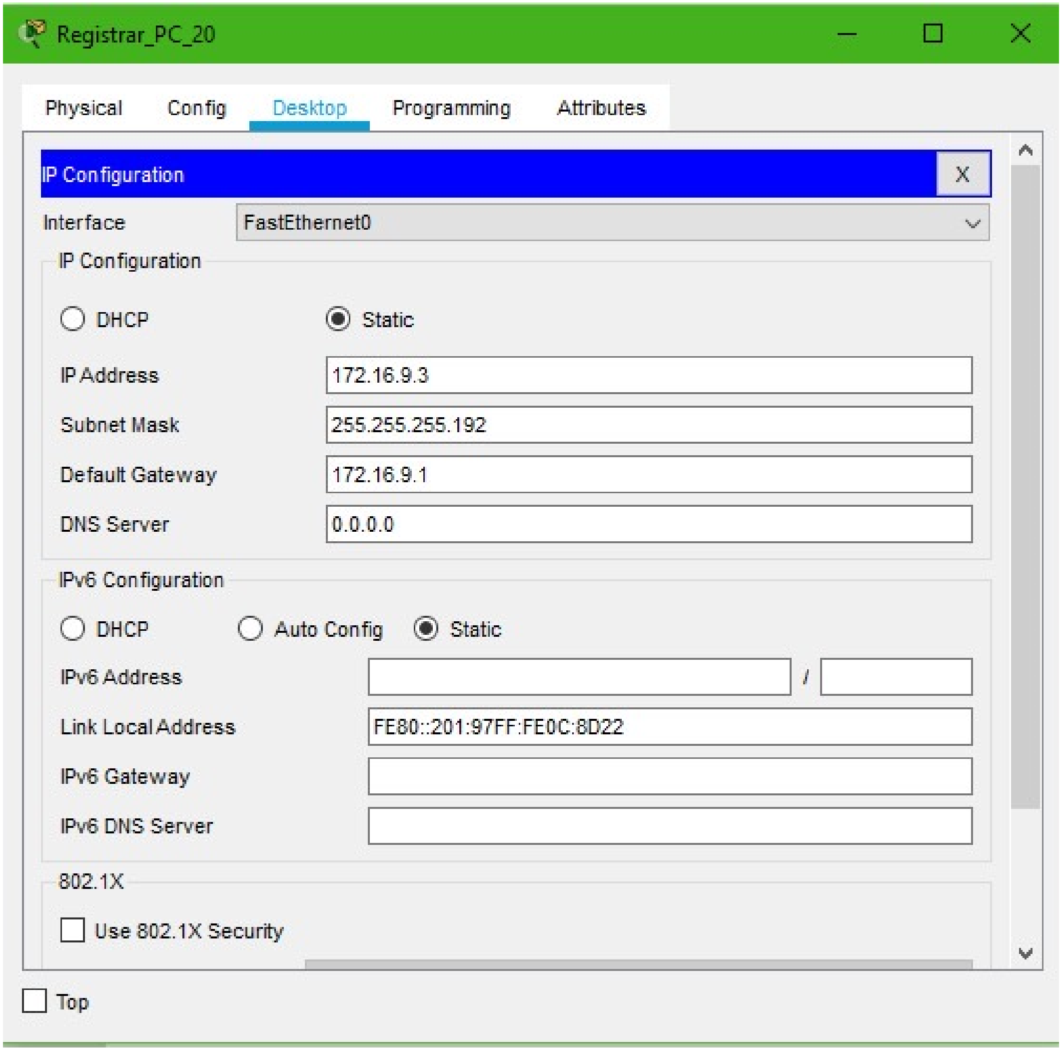

Figure 3.15 — Registrar department PC configuration

3.11 Faculty Department Configuration

Because the Faculty department was modelled with a much larger user base (250 devices), its end devices were configured to obtain their addressing automatically from a DHCP scope rather than by hand.

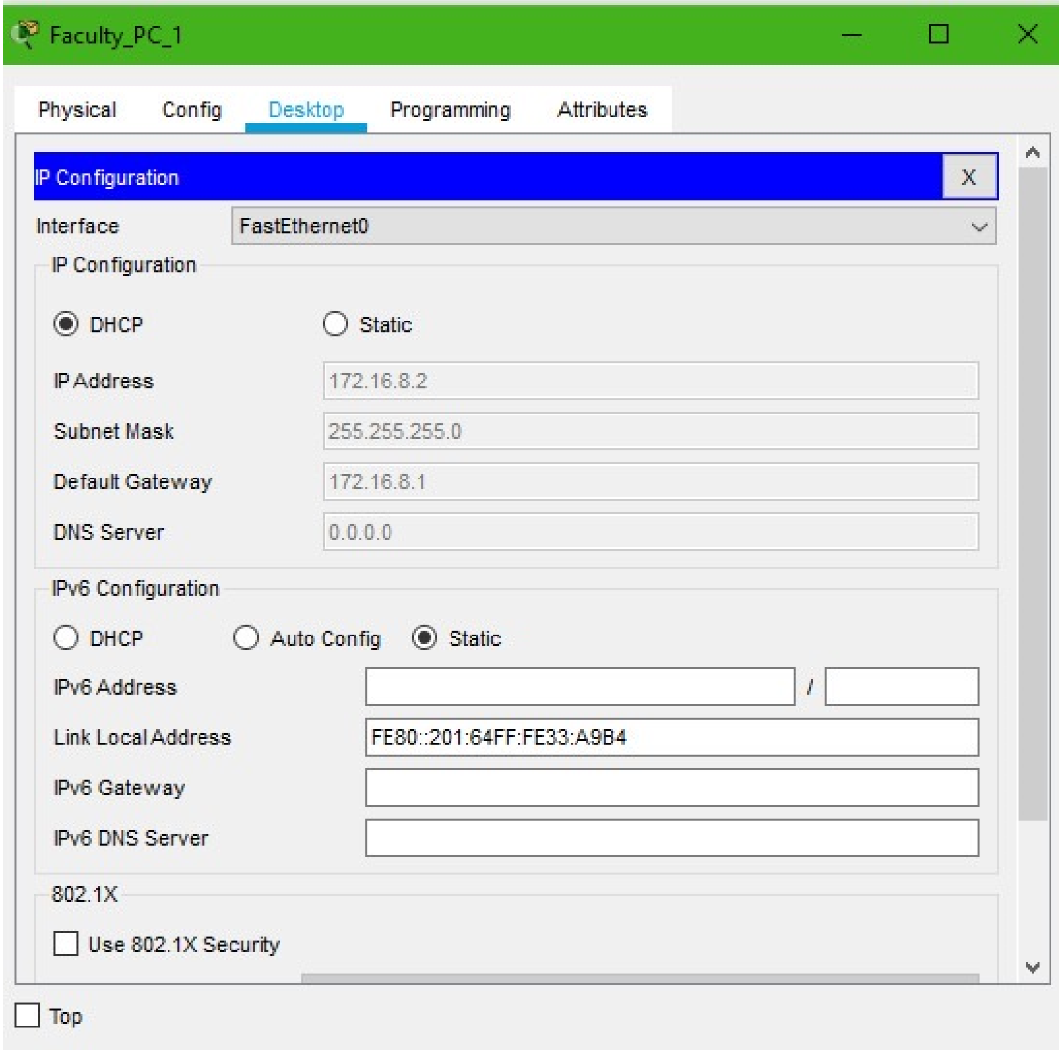

Figure 3.16 — Faculty department PC configuration (DHCP)

3.12 Student Department Configuration

The Student department, modelled with the largest user base of all (2000 devices), was likewise configured for DHCP-based addressing to keep administration manageable at scale.

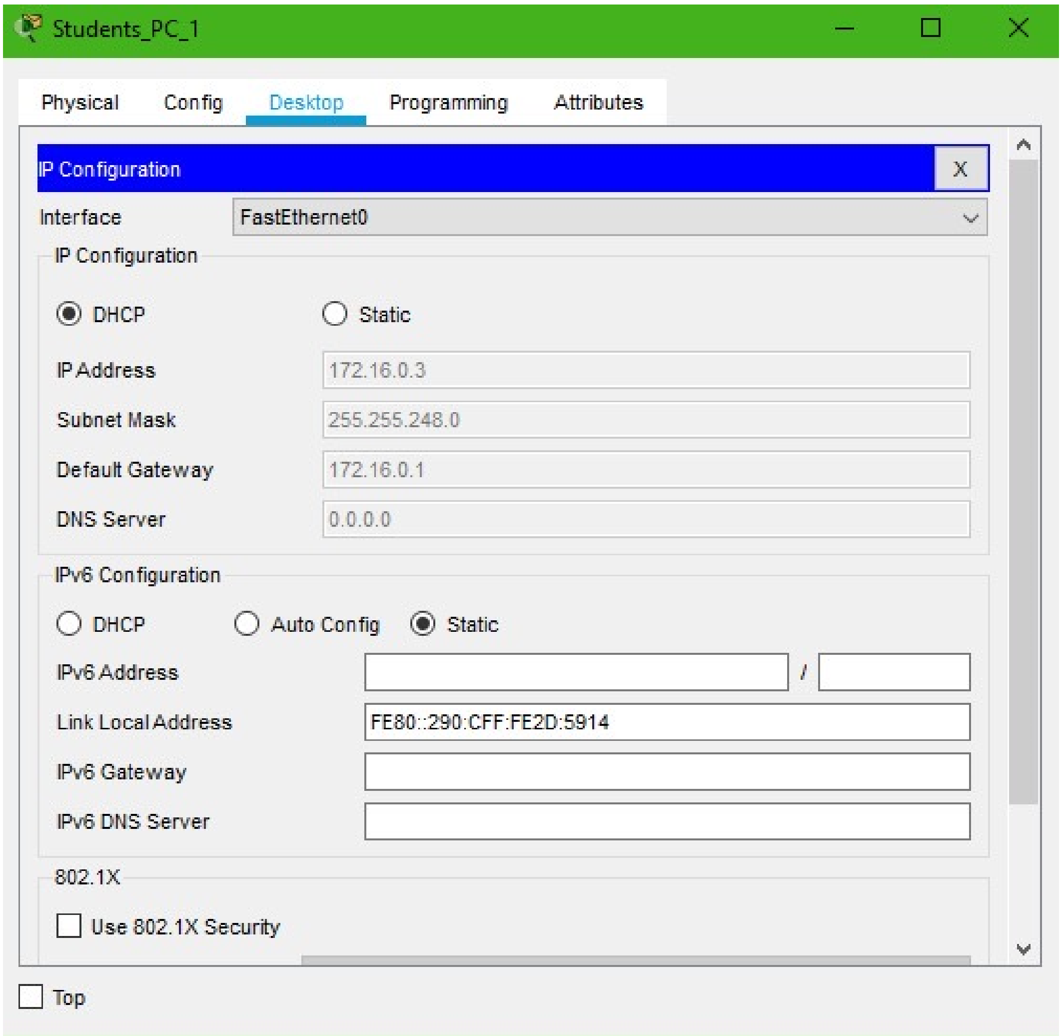

Figure 3.17 — Student department PC configuration (DHCP)

4. Results and Discussion

With every device and link configured, the network was validated in two complementary ways: connectivity (ping) tests between specific device pairs, and simulated PDU transmissions traced hop-by-hop through the topology.

4.1 Connectivity (Ping) Tests

A ping test measures round-trip latency and confirms basic reachability between two hosts, which makes it a quick first check of whether a routing or VLAN configuration is correct. Ping tests were run between department pairs in the same building, between department pairs across the two buildings, and between the external servers and campus PCs.

4.1.1 IT Department to Other Admin-Building Departments

The first pair tested was an IT department PC pinging an Accounts department PC, both within the Administration building.

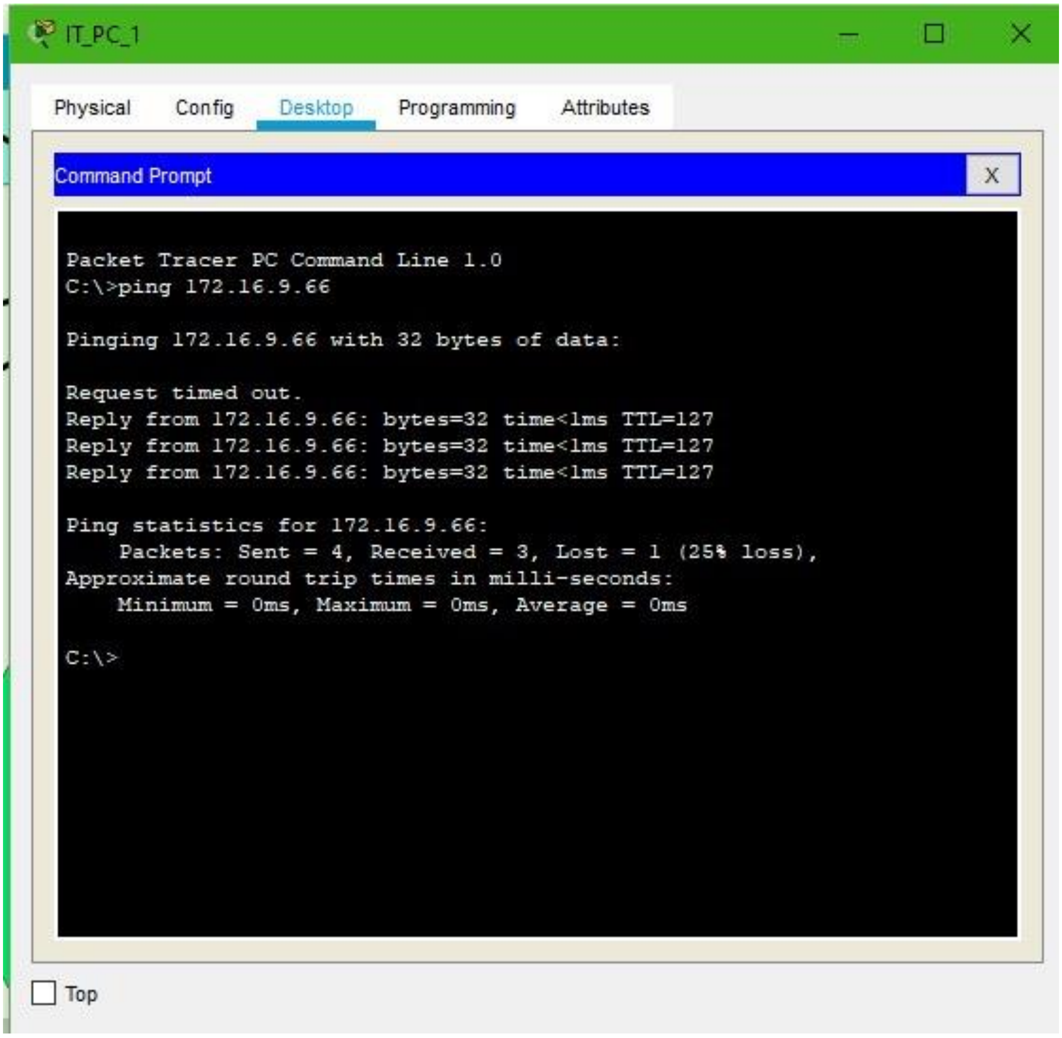

Figure 4.1 — Ping test: IT department to Accounts department

The same IT department PC was then used to ping a Registrar department PC.

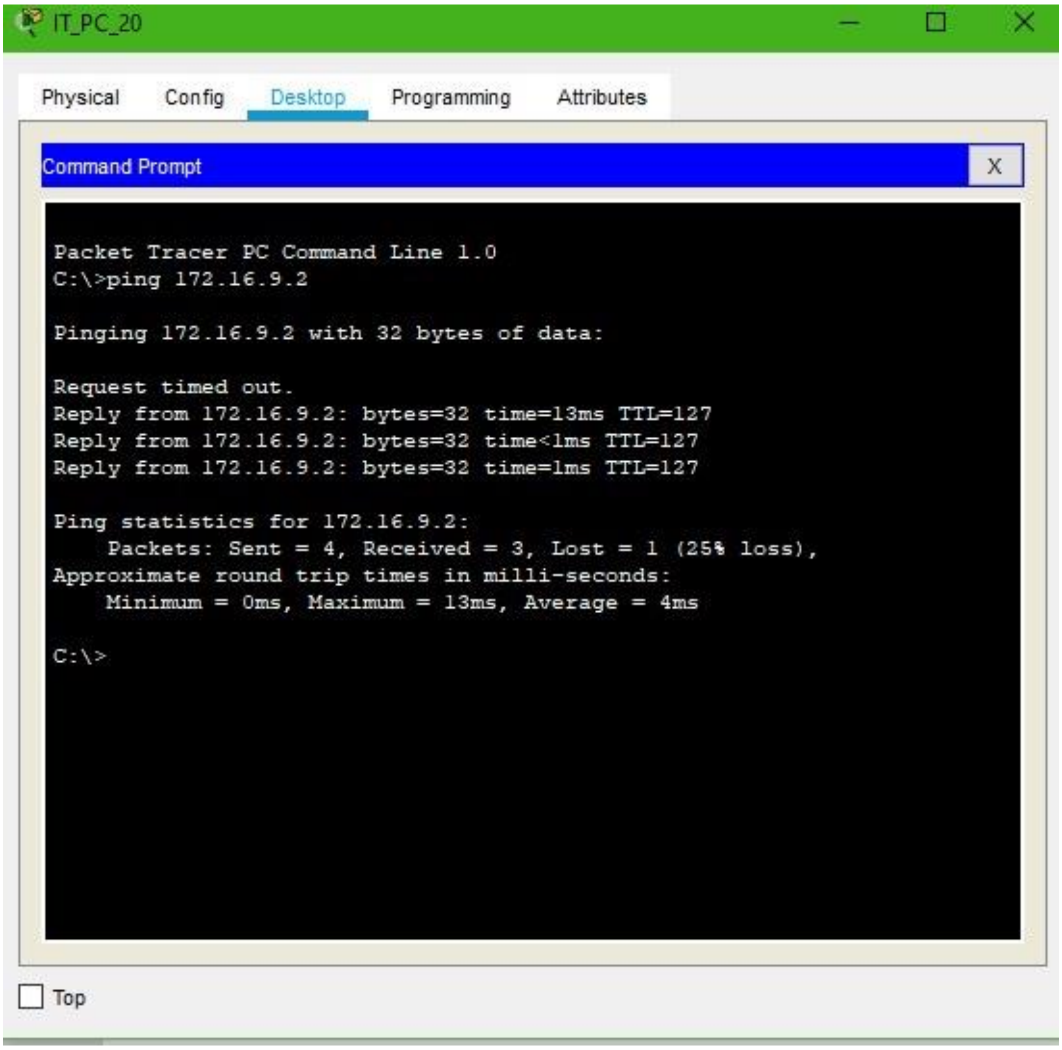

Figure 4.2 — Ping test: IT department to Registrar department

4.1.2 Accounts Department to Academic-Block Departments

Next, connectivity was tested across the two buildings: an Accounts department PC pinging a Student department PC in the Academic block.

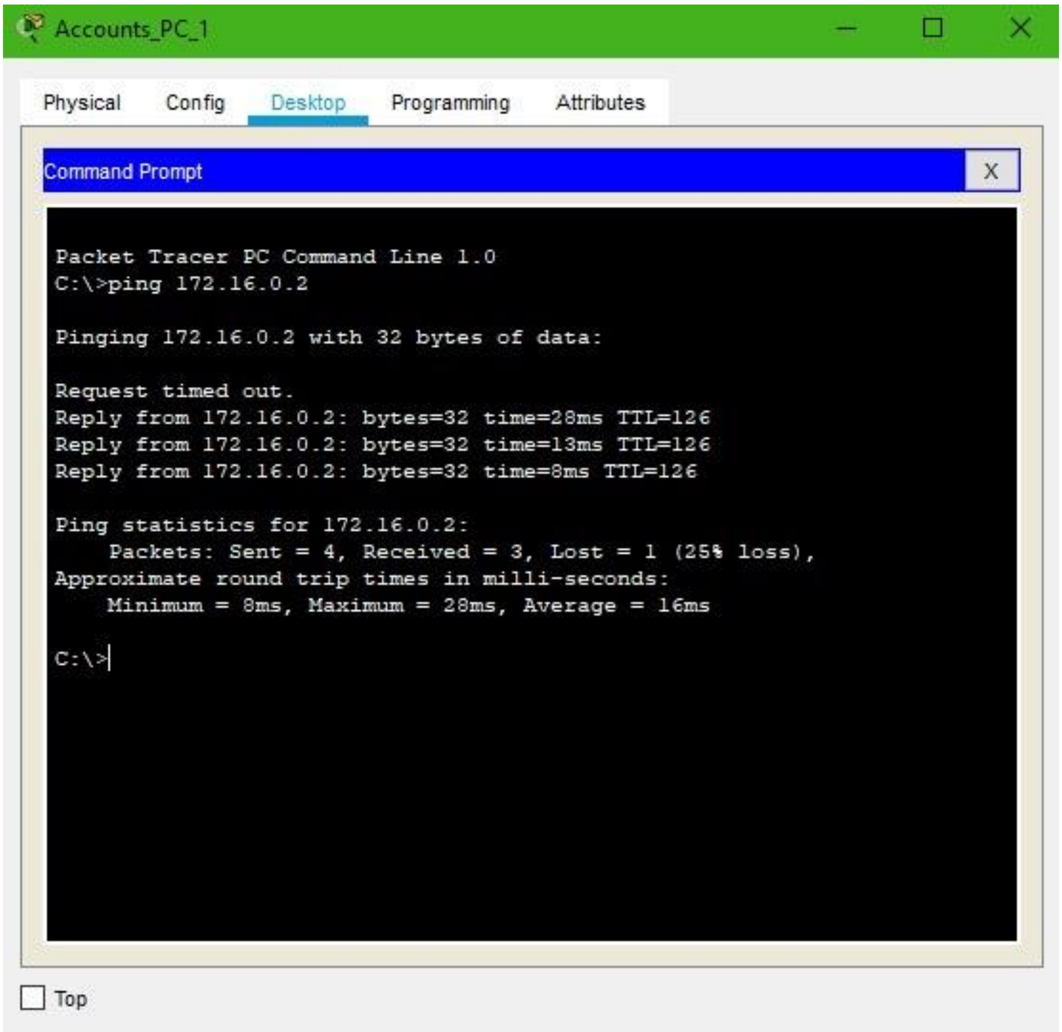

Figure 4.3 — Ping test: Accounts department to Student department

The same Accounts department PC was then used to ping a Faculty department PC.

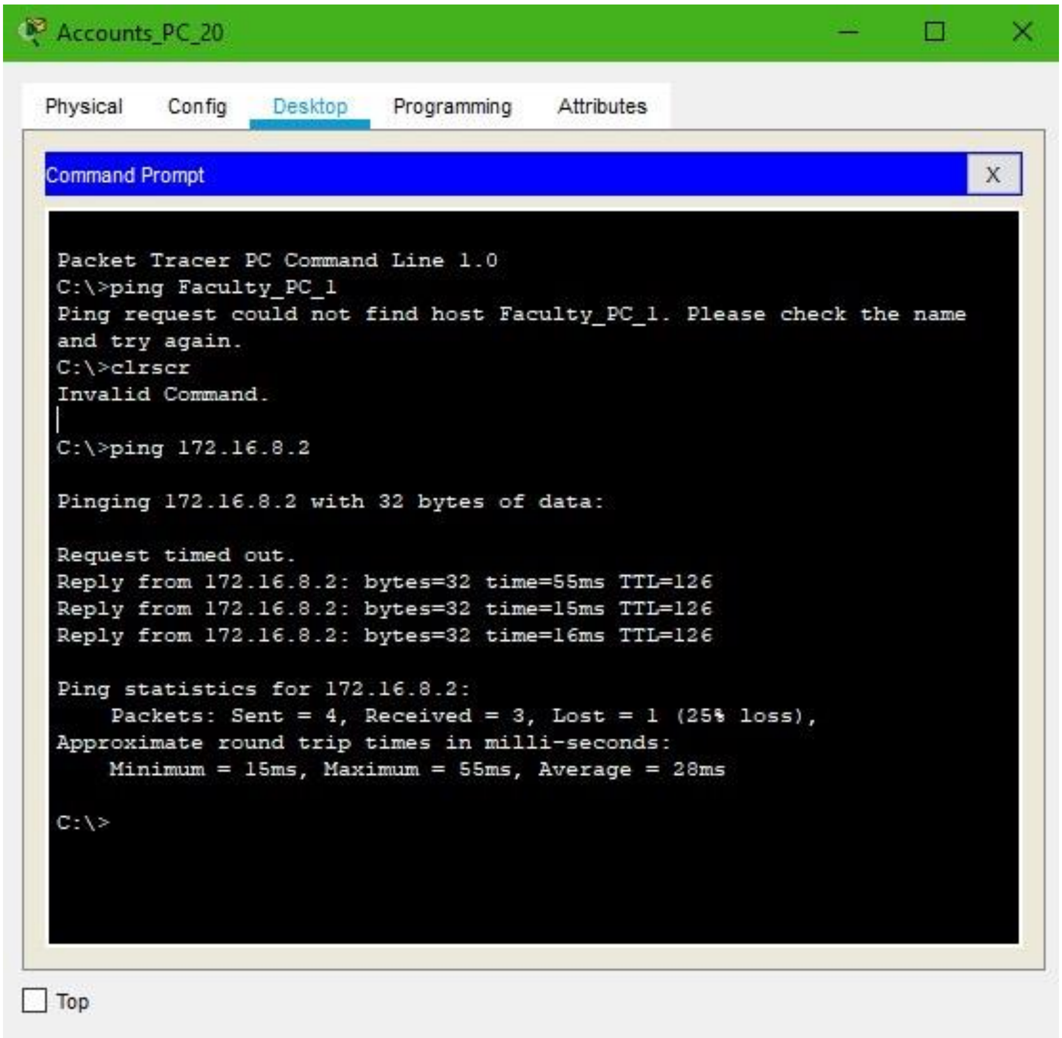

Figure 4.4 — Ping test: Accounts department to Faculty department

4.1.3 External Servers to Campus PCs

Finally, the two external servers were tested against PCs in both buildings, starting with the YouTube server pinging an Accounts department PC.

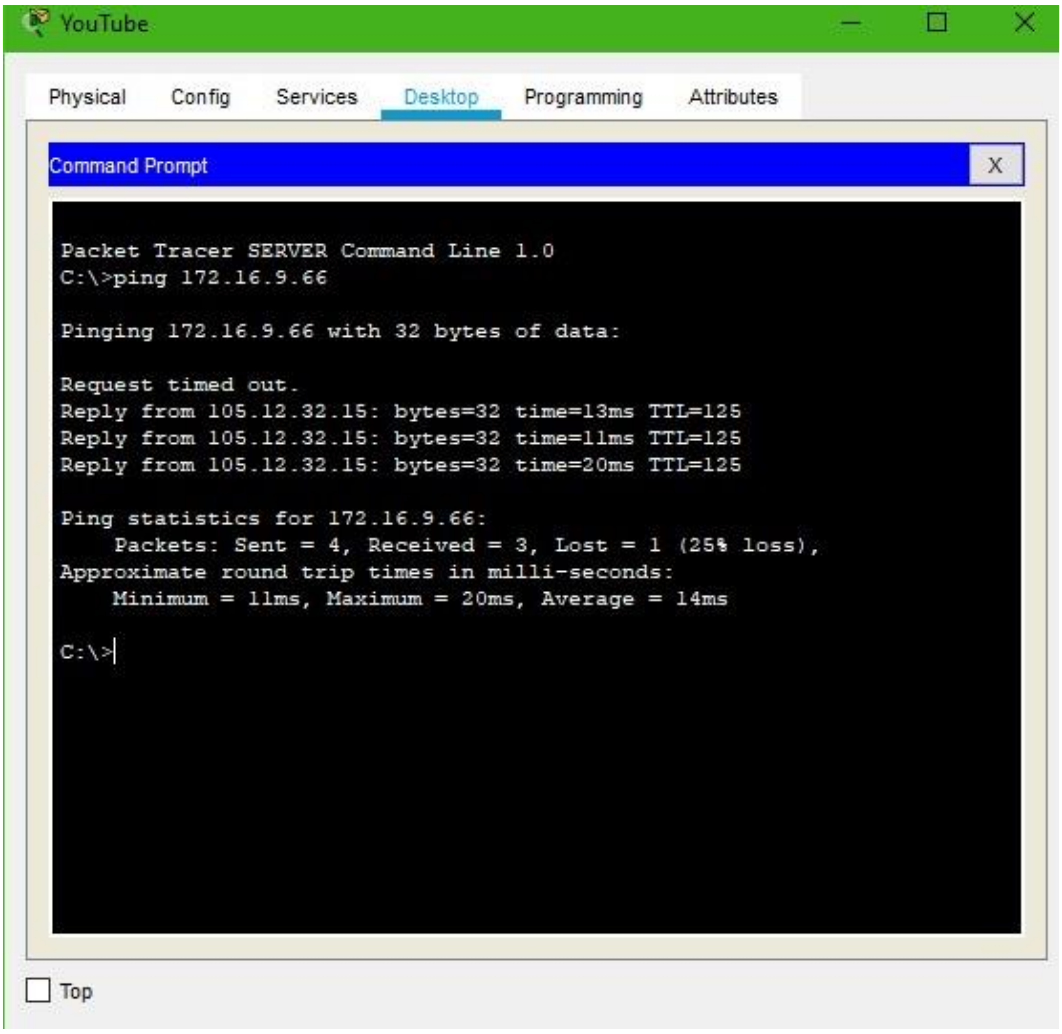

Figure 4.5 — Ping test: YouTube server to Accounts department

The Google server was then used to ping a Student department PC in the Academic block.

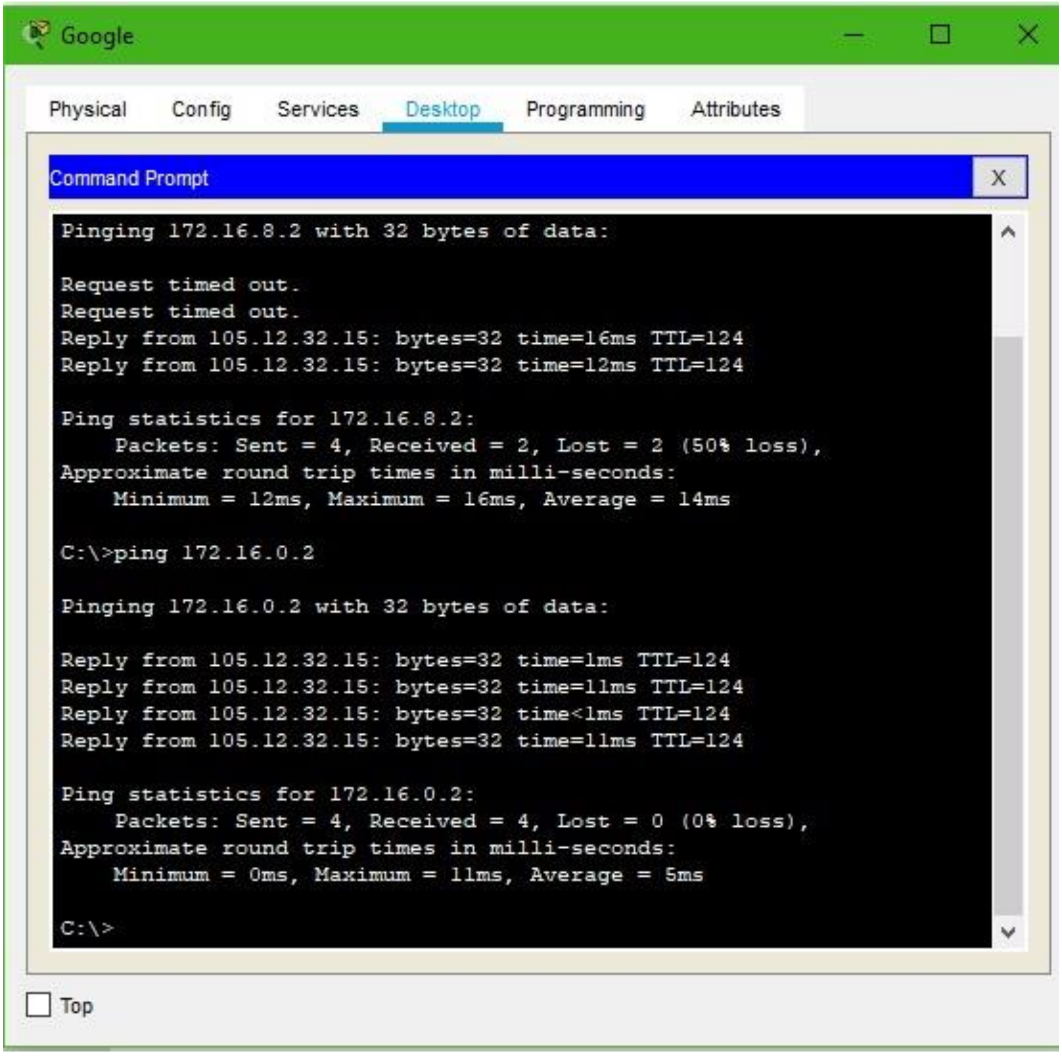

Figure 4.6 — Ping test: Google server to Student department

Every ping test above completed successfully, with all (or nearly all) of the packets delivered in each exchange and no test losing more than a single packet. This confirms that VLAN segmentation, inter-VLAN routing and the DHCP/static addressing scheme are all functioning together correctly across both buildings.

4.2 Packet Delivery Simulation

To look beyond simple reachability, Packet Tracer's Simulation mode was used to trace a single Protocol Data Unit (PDU) hop-by-hop as it moved between two chosen PCs, first within the same subnet and then across two different subnets.

4.2.1 PDU Simulation — Same Network

The first simulation sent a PDU between two PCs belonging to the same VLAN, so the frame stayed entirely at the switching layer without needing to be routed.

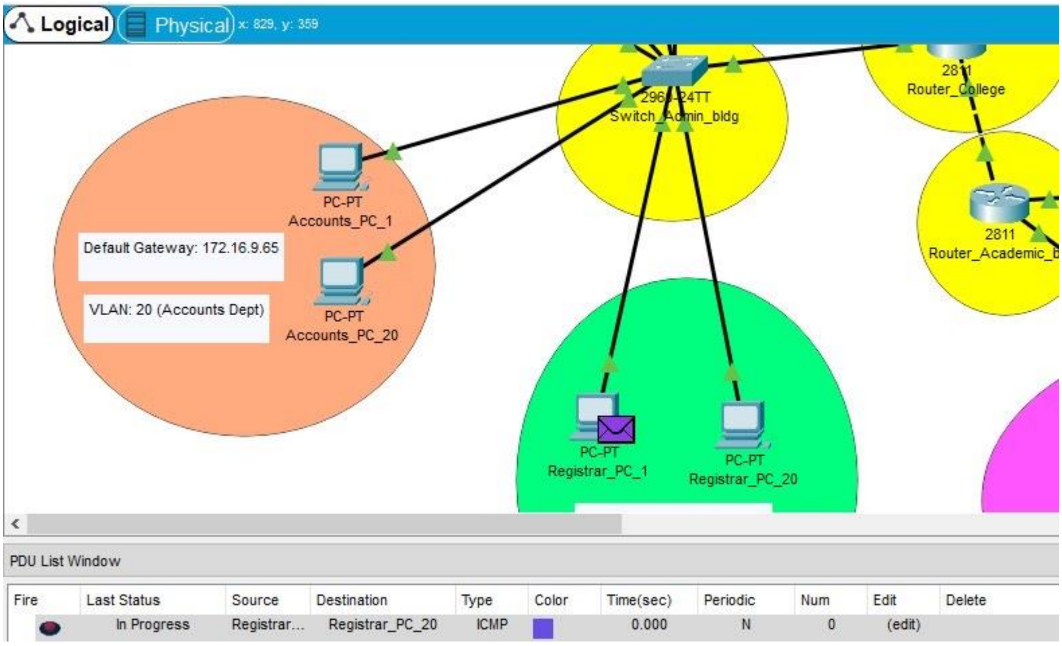

Figure 4.7 — PDU simulation between two PCs in the same network

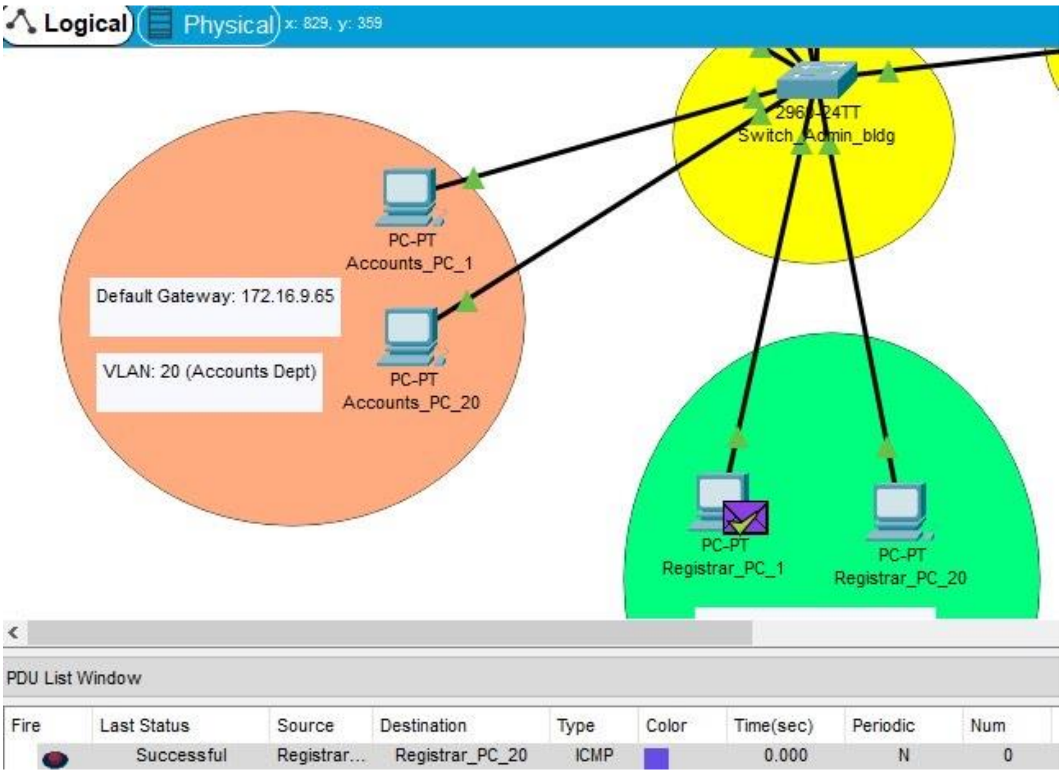

Figure 4.8 — PDU simulation result (same network)

4.2.2 PDU Simulation — Different Networks

The second simulation sent a PDU between two PCs in different VLANs, requiring the packet to be routed through the campus router (and, depending on the pair chosen, the border and ISP routers as well).

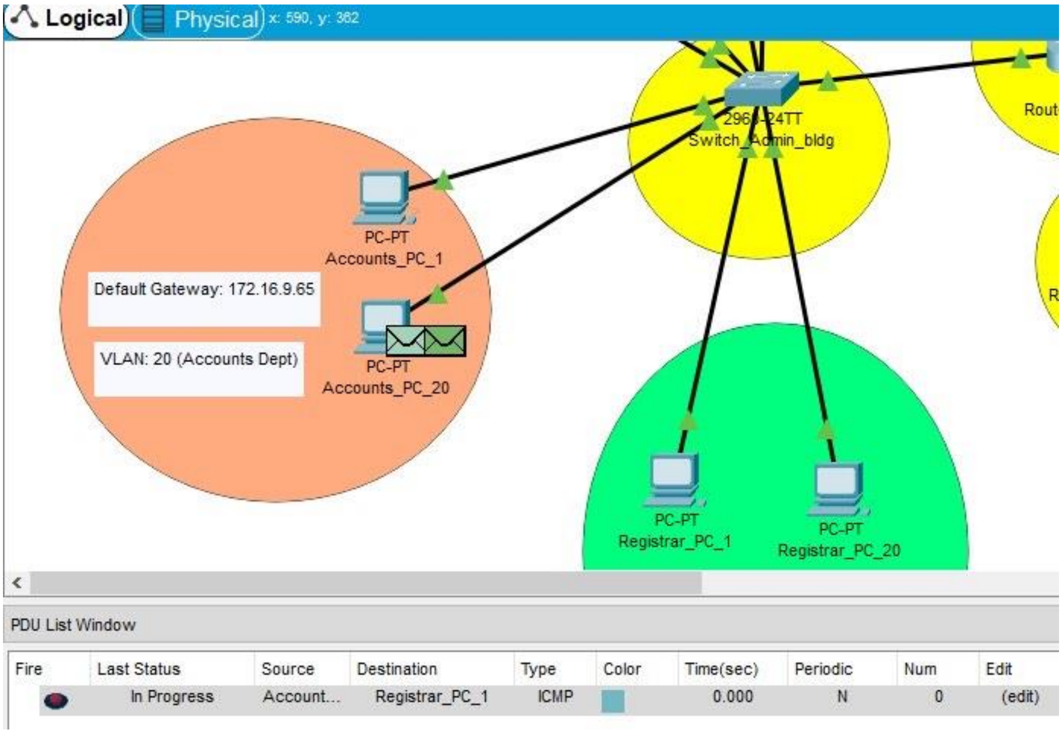

Figure 4.9 — PDU simulation between two PCs in different networks

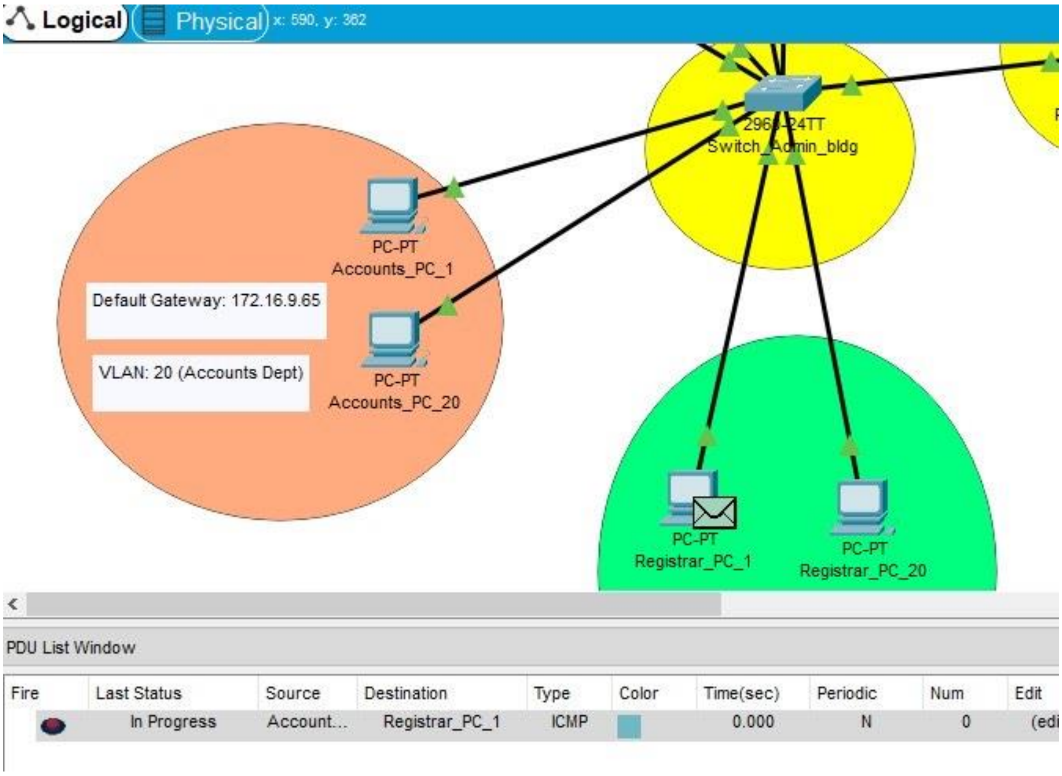

Figure 4.10 — PDU simulation, routing path detail

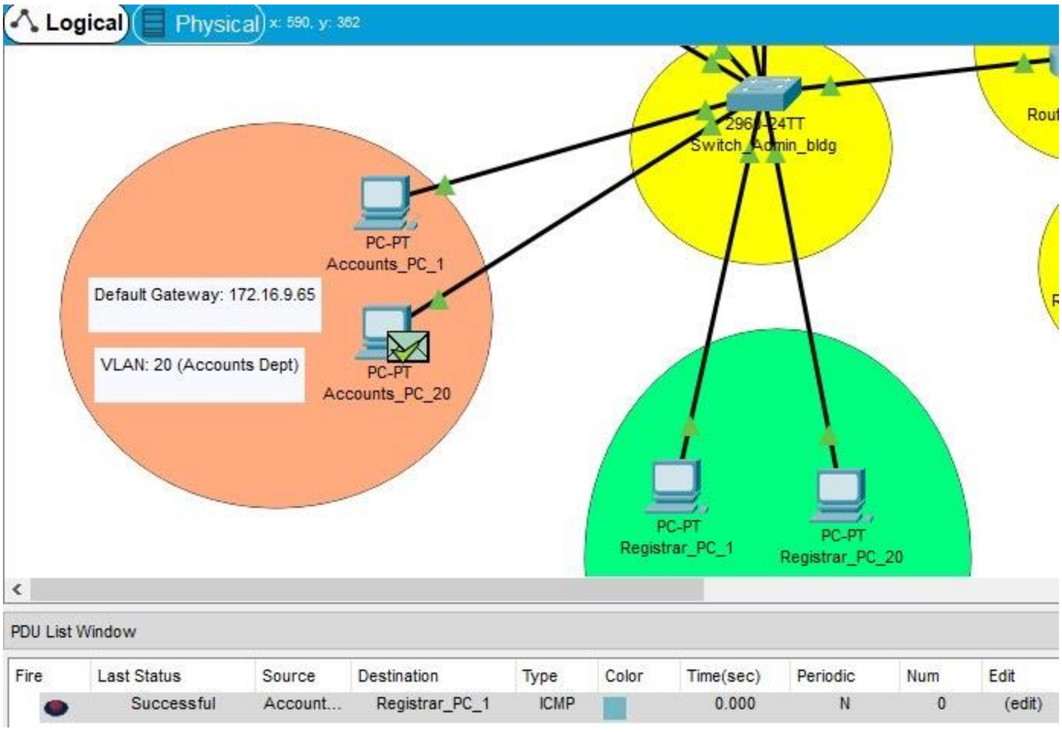

Figure 4.11 — PDU simulation result (different networks)

4.3 Discussion

Taken together, the ping tests and PDU simulations show that the design achieves its core objective: any two authorized devices on the campus, whether in the same department, in different departments of the same building, in different buildings, or on an external server, can successfully exchange data. The VLAN boundaries did not need to be relaxed at any point to make this work, which indicates that the inter-VLAN routing configuration on the campus router is correctly forwarding only the traffic it is meant to forward rather than accidentally bridging VLANs together.

The choice to use DHCP only for the Faculty and Student departments, while keeping IT, Accounts and Registrar on static addressing, also proved to be a reasonable trade-off in practice: the smaller departments retain tighter administrative control over their address assignments, while the two large departments avoid the overhead of manual configuration at a scale of hundreds or thousands of devices. One limitation of the current build is that only two representative end devices were modelled per department; a production deployment would need every physical host addressed, though the underlying VLAN and routing structure would not need to change to support that.

5. Conclusion and Future Work

5.1 Conclusion

This project designed, implemented and validated a five-department campus network for a medium-sized university using Cisco Packet Tracer. By combining VLAN-based segmentation, a structured private addressing plan, DHCP for high-density departments, and a layered router hierarchy leading out to an ISP, the network is able to keep departmental traffic isolated by default while still allowing every department controlled access to shared and external resources. The connectivity and PDU-simulation results confirm that the design is functionally correct and that it meets the reliability, security and efficiency goals set out at the start of the project. Because the design uses comparatively few routers and switches for the functionality it delivers, it also represents a cost-conscious approach that a real university with a similar department structure could realistically adopt.

5.2 Future Work

Several directions could extend this project further:

- Adding sub-departments as the university grows, each with its own VLAN, without disturbing the existing routing hierarchy.
- Scaling the Faculty and Student DHCP scopes (and adding a redundant DHCP server) to support continued growth in enrolment.
- Introducing access-control lists (ACLs) at the border router to enforce finer-grained security policies between departments and the Internet.
- Adding a wireless layer (WLAN) so that mobile devices can join departmental VLANs without a wired connection.
- Extending the simulation to model redundant links and failover, to test the network's resilience to a single link or device failure.

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